

GRAND RAPIDS, MB, Canada

WMO: 718580

Lat: 53.183N

Lon: 99.267W

Elev: 730

StdP: 14.31

Time Zone: -6.00 (NAC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-31.6	-26.8	-41.0	0.5	-29.1	-37.3	0.7	-26.1	17.0	2.8	15.2	0.9	3.2	300	0.651

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	22.0	83.6	66.8	80.1	65.0	77.0	63.5	69.4	79.0	67.4	76.3	65.6	73.7	8.3	250

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
66.2	99.1	73.5	64.2	92.2	71.8	62.4	86.5	70.0	34.1	79.2	32.4	76.3	30.9	73.8	80.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 16.6	14.4	12.9	-38.0	89.2	4.7	2.8	-41.4	91.2	-44.1	92.9	-46.8	94.5	-50.2	96.5		
(5)			-37.2	72.9	4.0	2.9	-40.1	75.0	-42.4	76.7	-44.6	78.3	-47.5	80.4		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	31.9	-2.4	1.5	14.5	30.9	43.6	56.5	63.5	61.1	51.3	37.6	19.6	3.7
	DBStd	25.49	14.78	13.81	14.70	11.35	8.56	7.93	5.90	6.34	8.23	8.03	12.18	14.62
	HDD50	7743	1623	1359	1100	577	237	25	0	2	81	393	911	1435
	HDD65	12177	2088	1779	1565	1024	665	273	101	155	416	850	1361	1900
	CDD50	1153	0	0	0	3	38	221	418	345	120	8	0	0
	CDD65	113	0	0	0	0	1	20	54	33	5	0	0	0
	CDH74	1388	0	0	0	0	47	300	593	372	75	1	0	0
	CDH80	315	0	0	0	0	7	92	126	80	10	0	0	0
	WSAvg	5.9	5.6	5.7	6.0	6.1	6.3	5.9	5.9	5.9	6.2	6.4	6.0	5.4
	PrecAvg	19.1	0.6	0.4	0.9	1.0	1.6	3.0	3.1	2.8	2.5	1.8	1.0	0.7
(15) Precipitation	PrecMax	27.6	1.4	0.9	2.0	3.1	4.3	7.8	7.6	5.5	4.9	3.9	2.5	1.3
	PrecMin	14.6	0.2	0.1	0.2	0.1	0.1	0.3	0.7	0.6	0.7	0.2	0.1	0.2
	PrecStd	3.4	0.4	0.2	0.5	0.7	1.0	1.7	1.8	1.5	1.0	1.1	0.6	0.4
	PrecStd	3.4	0.4	0.2	0.5	0.7	1.0	1.7	1.8	1.5	1.0	1.1	0.6	0.4
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	36.9	39.1	51.9	67.7	79.5	90.9	87.9	86.7	81.2	66.6	49.4	37.6
		MCWB	33.8	34.6	41.5	50.2	57.6	71.4	69.8	69.3	65.9	55.2	42.6	33.4
	2%	DB	30.6	33.2	46.3	59.4	72.7	81.5	83.5	82.0	75.5	59.5	43.4	31.4
		MCWB	28.6	30.6	38.6	46.5	54.8	64.7	67.0	67.3	62.1	50.7	38.6	29.1
	5%	DB	25.5	29.1	40.4	53.0	66.9	76.7	80.2	78.1	70.8	54.9	39.2	28.1
		MCWB	24.1	26.7	34.9	42.4	51.8	61.5	65.5	65.2	60.0	48.2	36.0	26.4
	10%	DB	20.9	23.9	35.4	47.5	60.7	72.4	77.2	74.6	66.2	50.9	35.1	24.3
		MCWB	19.9	21.9	32.1	39.6	49.1	60.0	64.4	63.7	57.8	46.0	32.9	22.8
	0.4%	WB	33.9	34.9	42.3	51.4	60.5	70.8	73.6	71.2	67.2	56.3	43.1	33.3
		MCDB	36.2	37.8	50.7	65.1	75.0	82.9	82.2	82.0	77.2	63.4	48.0	37.0
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	28.5	31.1	38.5	46.8	56.1	66.2	69.8	69.1	63.6	52.1	39.0	29.4
		MCDB	30.3	33.3	45.5	57.9	69.5	76.3	78.6	78.5	72.3	58.0	42.6	31.2
	5%	WB	24.1	26.7	35.1	43.3	52.9	63.6	67.7	67.2	61.0	48.9	36.0	26.5
		MCDB	25.5	29.4	39.6	51.9	64.2	72.8	76.9	75.2	68.8	53.8	39.0	28.2
	10%	WB	19.9	21.9	32.4	39.9	50.3	61.2	66.0	65.2	58.3	46.3	33.1	23.0
		MCDB	21.0	23.7	35.6	46.9	59.9	69.7	74.3	72.5	65.4	50.6	35.0	24.4
(35) Mean Daily Temperature Range	5% DB	MDBR	18.0	21.9	23.4	22.0	22.6	21.8	22.0	21.7	21.2	16.5	15.3	16.6
		MCDBR	19.0	21.7	26.6	29.9	35.8	29.1	26.5	26.5	27.9	24.4	17.7	16.4
		MCWBR	17.4	18.8	19.4	18.8	19.9	14.7	13.6	14.2	16.2	16.1	13.3	14.6
	5% WB	MCDBR	18.7	21.1	23.9	28.7	30.2	24.7	23.5	23.2	25.3	21.7	16.2	16.4
		MCWBR	17.1	18.4	17.5	18.7	18.0	13.9	13.4	13.6	15.7	15.1	12.4	14.7
(40) Clear-Sky Solar Irradiance	taub		0.252	0.253	0.277	0.343	0.352	0.379	0.391	0.366	0.330	0.294	0.247	0.233
	taud		2.221	2.283	2.347	2.374	2.433	2.389	2.388	2.464	2.546	2.577	2.401	2.255
	Ebn at Noon		233	276	292	282	285	277	271	272	271	261	241	221
	Edh at Noon		20	26	30	34	34	36	35	31	25	20	16	16
	RadAvg		269	580	1002	1438	1773	1824	1828	1490	1039	565	272	199
(45) All-Sky Solar Radiation	RadStd		31	56	81	146	162	131	92	115	110	71	34	24

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	-1.21	-1.06	N/A	N/A	N/A	N/A
(47) Regional (7 neighbors)	N/A	N/A	N/A	+2.92	N/A	N/A	-0.67	N/A	N/A	N/A	N/A

Nomenclature: See separate page